

Arun Krishnan

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Summary

With a solid foundation in Data governance, I have successfully transitioned into a Data Engineer role, specialising in ETL processes, database modelling, and optimising data pipelines for smoother data flow. I have a track record of enhancing data pipeline efficiency by as much as 30% and cutting processing time by 20%. Additionally, I have extensive in building web crawlers and overcoming blocking mechanisms. I am committed to leveraging data-driven insights to improve data management strategies.

Technical Skills

- **Programming Languages:** Python, SQL
- **Data Quality & Testing:** Unit Testing, Great Expectations(GE)
- **Container & Orchestration:** Docker, Azure Kubernetes, CI/CD, Container Registry
- **Data Science & Machine Learning:** Pandas, Numpy, Scikit-Learn, Tensorflow, Matplotlib
- **Databases & Storage:** PostgreSQL, Elasticsearch, StarRocks DB, Nessie (Data Lakehouse Catalog)
- **Big Data & Framework:** Apache Spark (PySpark), Apache Flink, Hive, Kafka, Apache Airflow, Debezium
- **Cloud Platform & ETL Services:** ADLS Gen2, Azure Data Factory, Azure Event Hub, Azure Fabric, Synapse, Azure Databricks, AWS S3

Experience

AEDGE AICC India Pvt. Ltd. (Armada.ai)

June 2025 - Present

L2 Data Engineer

Trivandrum

- Designed and implemented a real-time data pipeline from PostgreSQL to Azure Synapse, enabling 100% CDC using Debezium and Azure Event Hub.
- Developed Spark jobs on Databricks to process streaming CDC data and load it into Synapse Data Warehouse with reducing data latency by 90%.
- Delivered an ML forecasting pipeline on Databricks, posting daily predictions to a REST API, improving forecast delivery speed by 80%.
- Orchestrated end-to-end workflows using Airflow, increasing pipeline reliability by 95% with automated retries and alerting.
- Deployed Debezium and Airflow on AKS, improving infrastructure scalability and reducing manual maintenance by 70%
- Collaborated cross-functionally with Backend, AI/ML, Power BI, and Frontend teams to deliver integrated data solutions that drive business insights and product functionality.

Turbolab Technologies Pvt. Ltd. (Scrapehero)

July 2022 - June 2025

Associate Data Engineer

Kochi

- Architecting scalable data infrastructures using PySpark, Dremio, and Iceberg to handle large volumes of data, achieving a scalability improvement of up to 40%.
- Designing and implementing efficient ETL processes, reducing data processing time by 25% and enhancing data quality by 20%
- Developing and optimising data models and schemas using dimensional modelling for effective data storage and retrieval, improving data access speed by 30%.
- Collaborated with cross-functional teams to integrate machine learning models into the data pipeline, increasing predictive analytics accuracy by 25%.
- Implementing data governance policies and procedures to ensure data security, compliance, and privacy, achieving a 95% data accuracy rate.
- Implement Great Expectations(GE) to improve data quality by 67%, utilising data profiling, validation, and automated monitoring to ensure adherence to essential quality metrics like completeness, uniqueness, and format accuracy.

Core Competencies

- Analytical
- Strategic
- Mentoring
- Research
- Innovative
- Decisiveness
- Debugging
- Documentation

Projects

- **Data Ingestion Pipeline PostgreSQL to Synapse Armada (Professional)**
 - Built a real-time ingestion pipeline capturing 100% of CDC changes from PostgreSQL via Debezium, streaming 1M events/day through Azure Event Hub, processed with Spark on Databricks, and loaded into Azure Synapse with 90% reduction in data latency. Achieved 99.9% pipeline uptime with automated recovery and monitoring, ensuring reliable and timely data delivery for downstream analytics.
 - **Technologies:** PostgreSQL, Debezium, Event Hub, Spark, Databricks, Delta Lake, Synapse, AKS.
- **Integration of ML Model with POST API using Databricks Armada (Professional)**
 - Integrated a time-series forecasting ML model in Databricks to generate daily predictions, posting results to a secured POST API, reducing manual reporting by 85%. Automated the end-to-end pipeline using Apache Airflow, improving model delivery speed and reliability by 90%
 - **Technologies:** Databricks, PySpark, MLflow, REST API, Apache Airflow, Kubernetes (AKS), Python
- **E-commerce Data Streaming ETL Pipeline Turbolab (Professional)**
 - Designed a scalable architecture of a pipeline that consume data from Kafka(source) using PySpark as RDD, it improve 45% faster processing capability compared to normal data frame. Then Transform the data for compact to the iceberg and Postgres table, then process the data under quality validation so it can improve data quality by 67%. The it write it into Iceberg data lake and Postgres table(Destination)
 - **Technologies:** Python, PySpark, Iceberg, Nessie, Postgres, Shell Script, Great Expectation(GE)
- **Product's Customer Reviews Data Batch Processing ELT Pipeline Turbolab (Professional)**
 - Streamlined data ingestion from Kafka into a data lake, enforced data quality checks using GE in Airflow, and orchestrated a Spark batch pipeline to implement transformations and aggregations, ensuring clean data is delivered to a Postgres database with 100% assurance that the data has successfully reached the Postgres DB.
 - **Technologies:** PySpark, Great Expectation(GE), Iceberg, Airflow, LLM, NLP, Python, Postgres
- **Custom Data Quality Rules in GE with PySpark Turbolab (Professional)**
 - When it comes to the stream pipeline, the QA is completed by the time of streaming. While working on the pipeline with Spark, certain fields have specific rules and conditions. In this case, we customise the Great Expectations (GE) tools by modifying the source code and implementing custom rules, which makes the end-to-end data processing 72% faster.
 - **Technologies:** PySpark, Great Expectation(GE), Python
- **Pipeline Managing Alert System Turbolab (Professional)**
 - To monitor the pipeline and alert the team when there are discrepancies, we will implement a complete Python-based project that can be configured using a YAML file. This file will be included in the project repository. The system will scan the project to locate the YAML file and monitor the pipeline using the configurations specified within it. This approach helps to reduce pipeline downtime by 83% and improve availability by 67%.
 - **Technologies:** Python, Google Chat, API, Crone

Education

SHM Engineering College, Kadakkal
Computer Science and Engineering

March 2016 - Jun 2020
CGPA: 6.83

Classic ITI, Anchal
Computer Operation & Programming Assistant

April 2014 - May 2015
MARKS: 72%